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In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.optimize import minimize
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In [2]: # Read data
df = pd.read_csv("Advertising.csv")
dfs = df.iloc[:5,] # Subset first 5 rows
dfs
```

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Out[2]:
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	Unnamed: 0	TV	Radio	Newspaper	Sales
0	1	230.1	37.8	69.2	22.1
1	2	44.5	39.3	45.1	10.4
2	3	17.2	45.9	69.3	9.3
3	4	151.5	41.3	58.5	18.5
4	5	180.8	10.8	58.4	12.9

```
In [3]: def obj(w):
    m1, m2, m3, b = w
    tv = dfs.loc[:, 'TV'].values
    radio = dfs.loc[:, 'Radio'].values
    news = dfs.loc[:, 'Newspaper'].values
    y = dfs.loc[:, 'Sales'].values
    pred_Sales = (m1 * tv) + (m2 * radio) + (m3 * news) + b

    return ((pred_Sales - y) ** 2).mean()
```

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In [4]: # Calculuate values to minimize distance to best fit
res = minimize(obj, x0 = [0, 0, 0, 0])
res.x
```

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Out[4]: array([ 0.07054428,  0.23961653, -0.04017871, -0.1419681 ])
```

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In [5]: # Prepare data for plotting
optimized_m1, optimized_m2, optimized_m3, optimized_b = res.x
tv_data = dfs['TV'].values
radio_data = dfs['Radio'].values
news_data = dfs['Newspaper'].values
actual_sales = dfs['Sales'].values
predicted_sales = (optimized_m1 * tv_data) + (optimized_m2 * radio_data) + (optimiz
predicted_sales
```

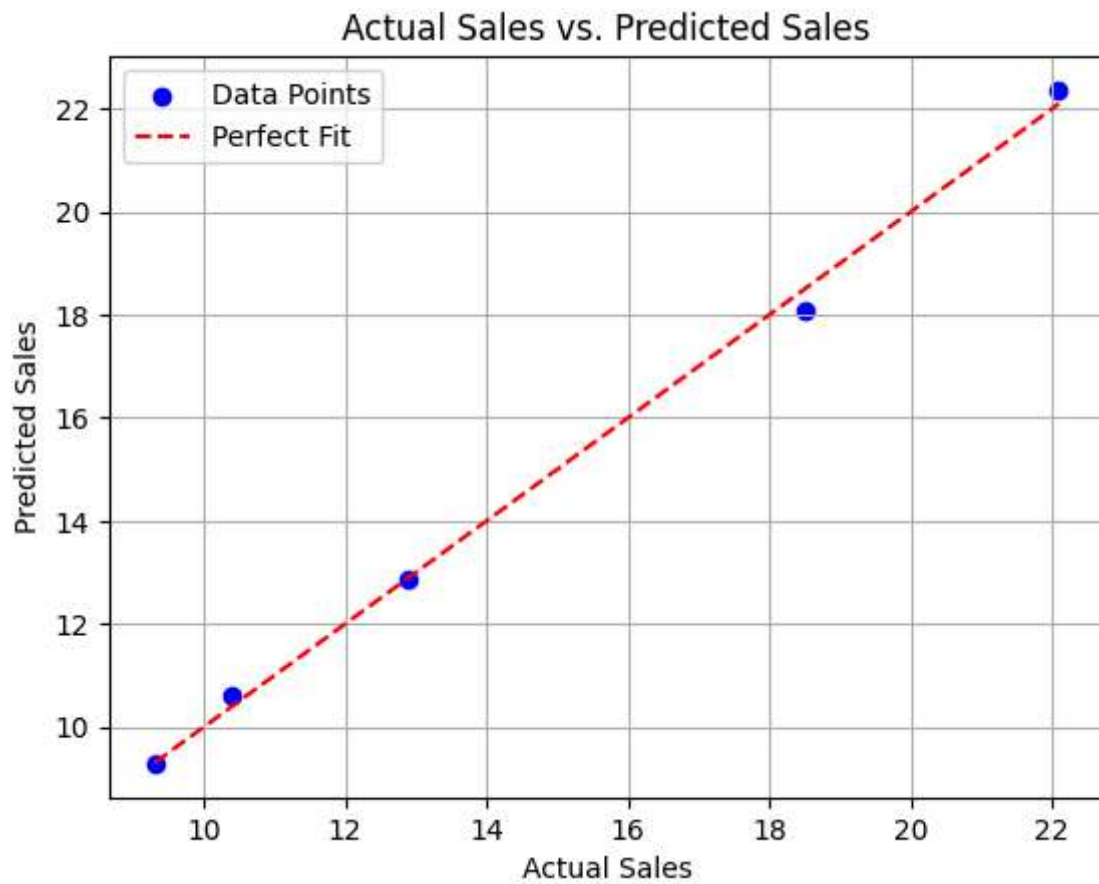
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Out[5]: array([22.36741001, 10.60212241,  9.28540776, 18.09119926, 12.85386052])
```

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In [6]: plt.scatter(actual_sales, predicted_sales, color='blue', label='Data Points')

perfect_line = np.linspace(min(actual_sales), max(actual_sales), 100) # return even
```

```
plt.plot(perfect_line, perfect_line, color='red', linestyle='--', label='Perfect Fi

plt.title('Actual Sales vs. Predicted Sales')
plt.xlabel('Actual Sales')
plt.ylabel('Predicted Sales')
plt.legend()
plt.grid(True)
plt.show()
```



In []: